📘 Data Cleaning in Power BI

🔹 Load Your Dataset

Steps:

1. Open **Power BI Desktop**
2. Click **Home tab** → **Get Data** → **Excel**
3. Select your dataset file
4. Click **Transform Data**

➡️ Power Query Editor opens

🔹 1. Remove Duplicates

✅ Example 1:

Column: **Customer ID**

Steps:

1. Go to Power Query Editor
2. Locate column **Customer ID**
3. Right-click on **Customer ID** column header
4. Click **Remove Duplicates**

✅ Example 2:

Remove duplicate rows (all columns)

Steps:

1. Press **Ctrl + A** to select all columns
2. Right-click on any column header
3. Click **Remove Duplicates**

🔹 2. Handle Missing Values

✅ Example 1:

Column: **Sales** (null values)

Steps:

1. Select column **Sales**
2. Right-click on **Sales**
3. Click **Replace Values**
4. In dialog box:
   - Value to Find → `null`
   - Replace With → `0`
5. Click **OK**

✅ Example 2:

Remove blank rows

Steps:

1. Go to **Home tab**
2. Click **Remove Rows**
3. Select **Remove Blank Rows**

🔹 3. Change Data Types

✅ Example 1:

Column: **Sales** (Text → Number)

Steps:

1. Select column **Sales**
2. Right-click on column header
3. Click **Change Type** → **Whole Number**

✅ Example 2:

Column: **Order Date** (Text → Date)

Steps:

1. Select **Order Date**
2. Right-click
3. Click **Change Type** → **Date**

🔹 4. Remove Unnecessary Columns

✅ Example 1:

Column: **Temp Column**

Steps:

1. Right-click on **Temp Column**
2. Click **Remove**

✅ Example 2:

Keep only required columns

Steps:

1. Select required columns (Ctrl + click multiple columns)
2. Right-click on selected columns
3. Click **Remove Other Columns**

🔹 5. Rename Columns

✅ Example 1:

Cust_ID → Customer ID

Steps:

1. Right-click on **Cust_ID**
2. Click **Rename**
3. Type **Customer ID**
4. Press Enter

✅ Example 2:

Amt → Sales Amount

Steps:

1. Right-click on **Amt**
2. Click **Rename**
3. Enter **Sales Amount**

🔹 6. Trim & Clean Text

✅ Example 1:

Column: **Customer Name** (extra spaces)

Steps:

1. Select **Customer Name**
2. Go to **Transform tab**
3. Click **Format** → **Trim**

✅ Example 2:

Remove unwanted characters

Steps:

1. Select column
2. Go to **Transform tab**
3. Click **Format** → **Clean**

🔹 7. Split Columns

✅ Example 1:

Column: **Location** = "Bangalore, India"

Steps:

1. Select **Location**
2. Right-click on column header
3. Click **Split Column** → **By Delimiter**
4. Choose delimiter → **Comma (,)**
5. Click **OK**

✅ Example 2:

Column: **Full Name**

Steps:

1. Select **Full Name**

2. Right-click
3. Click **Split Column → By Delimiter**
4. Choose delimiter → **Space**
5. Click **OK**

🔹 8. Merge Columns

✅ Example 1:

Columns: **First Name + Last Name**

Steps:

1. Select **First Name** and **Last Name** (Ctrl + click)
2. Right-click on selected columns
3. Click **Merge Columns**
4. Choose separator → **Space**
5. Click **OK**

✅ Example 2:

Columns: **City + Country**

Steps:

1. Select both columns
2. Right-click
3. Click **Merge Columns**
4. Choose separator → **Comma or Space**

🔹 9. Replace Values

✅ Example 1:

Column: **City**

Replace: `Blr → Bangalore`

Steps:

1. Right-click on **City**
2. Click **Replace Values**

3. Enter:
   - Find → `Blr`
   - Replace → `Bangalore`
4. Click **OK**

✅ Example 2:

Column: **Gender**

Replace:

- `M → Male`
- `F → Female`

Steps:

1. Right-click on **Gender**
2. Click **Replace Values**
3. Replace one by one

🔹 10. Format Dates

✅ Example 1:

Column: **Order Date**

Steps:

1. Right-click on **Order Date**
2. Click **Change Type** → **Date**

✅ Example 2:

Extract Year from **Order Date**

Steps:

1. Select **Order Date**
2. Go to **Add Column tab**
3. Click **Date** → **Year** → **Year**

🔹 Final Step

Steps:

1. Click **Home tab**
2. Click **Close & Apply**

In [ ]:

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